

A Language Acquisition Support System that Presents Differences and Distances from Model Speech

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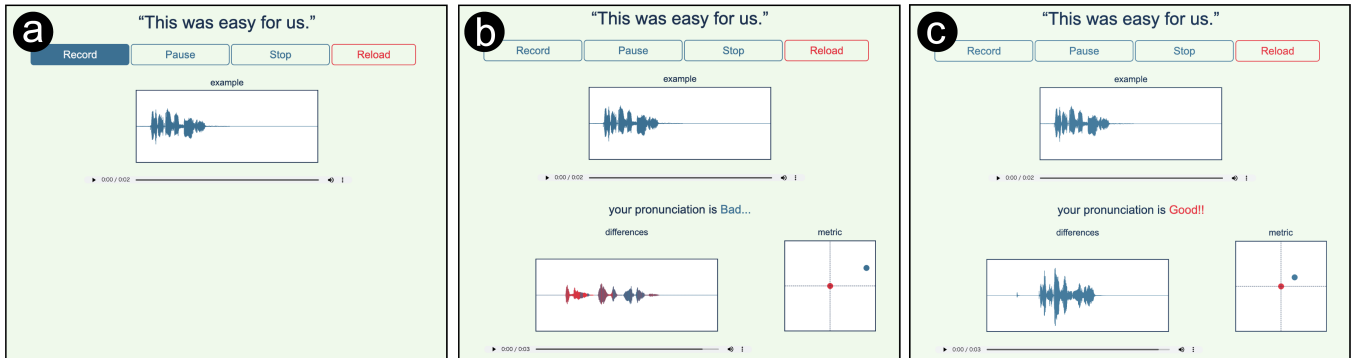


Figure 1: The flow of acquiring a language (English in this example): (a) First, a sentence of English to be practiced and a speech of a model (native speaker) are given. (b) In the next screen, the system analyzes the learner's English speech and draws a picture of how far away the user's speech is from the native speaker's speech and where it differs from the native speaker's speech. (c) The user repeats speaking to correct the different parts pointed out by the system, and finishes learning when the user's own speech is close to the native speaker's speech.

ABSTRACT

It is difficult for language learners to know whether they are speaking well and, if not, to know where their speech differs from that of native speakers and how much their speech differs from that of native speakers. Therefore, we propose a novel language learning system that solves these problems. The system uses self-supervised learning to determine whether the user's speech is good or not. The system also shows where the user's speech differs from the native speaker's speech by highlighting the corresponding places on the user's speech waveform. It also represents the learner's speech and the native speaker's speech as points on a two-dimensional coordinate to show the user how far apart they are. We expect that the learner's speech will gradually become better by repeatedly modifying the speech to eliminate these differences and bring the distance closer.

CCS CONCEPTS

• Human-centered computing → Visualization techniques.

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KEYWORDS

Language Acquisition, Self-supervised Learning, Attention Models, Neural Networks

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1 INTRODUCTION

Learning a new language that is not one's native tongue, especially learning to speak, is challenging because it differs in many aspects such as vocabulary, grammar, phonology, and accent. In order to learn a new language effectively, learners need to understand the differences between themselves and the native speakers of the language. The most common method is dictation, in which native speakers or experts evaluate how well the learner's speech is heard [2, 4]. Since this method requires a lot of time and effort, our research aims to automate this process. We propose a novel system that discriminates between the speech of native speakers and learners with a small amount of data by using self-supervised learning. In addition, this system helps the user to learn the language by visualizing what is different between the user's speech and the native speaker's speech, and how far apart the user's speech is from the native speaker's speech. The core model of the system is trained in

an end-to-end manner, so the system is capable of learning a variety of languages by preparing the data of the language the user wants to learn. This is in contrast to most existing language learning applications. For example, when a user wants to learn English, if they want to imitate the speech of native speakers of British English, they can build the system by collecting data of native speakers of British English. If they want to learn English which is widely used as a second language and is not limited to any country, they can build the system by collecting the data of English speakers around the world. The research on the development of computer-aided pronunciation detection systems has been actively conducted as part of CAPT (Computer-Aided Pronunciation Training) [8]. In these existing studies, the differences between the learner's speech and the native speaker's speech are calculated in detail in several aspects (intonation, rhythm, phonemes, etc.). However, our method does not show the differences between learners and native speakers in specific aspects such as these. This is because it does not necessarily mean that correcting these specific aspects makes the speaker's pronunciation intelligible. Instead, our system intuitively shows the differences and distances between the speaker's own speech and the model's, so that the speaker can learn the language through repeated interactions to eliminate the differences and bring the distances closer together. Another problem with existing methods is that they require two identical utterances for the same sentence, one from the learner of the target language and one from the native speaker. Our difference visualization method, on the other hand, simply needs the learner's and native speaker's speech data and does not require that the speeches match. In other words, the user is free to choose the sentences they want to learn utterances from, including sentences that are not included in the dataset.

2 OUR SYSTEM

Our system assists learners in learning a language that is not their mother tongue by showing the difference and distance between the learner's voice and that of the native speaker. Fig. 1 shows the flow of learning a language using the system. This figure shows an example of a non-native English speaker learning English.

First, as shown in Fig. 1(a), the system presents the sentence to be learned and the model utterance of the sentence (the utterance of one of the native speakers of the target language). The learner listens to this speech and checks how to pronounce the target sentence in a way that is easy for the listener to understand.

Next, the user utters the given sentence. As shown in Fig. 1(b), the system analyzes the user's utterance and presents the user with the differences and distances between the learner's utterance and the utterances of many speakers whose native language is the target language. As for the differences, in the waveform of the user's speech shown in the figure, the parts that differ from native speakers are indicated in red. The darker this red area is, the greater the difference is. Here, it is up to the learner to decide what groups to assume as native speakers of the target language. The difference here will be the difference from the average of the native speakers of the target language who were used to train the model. Therefore, if the user wants to learn speech that is close to American English, he or she can use the speech data of native speakers of American English as native speaker data. Alternatively, if the user wants to

learn what is widely used in English, the model can be trained using the speech data of English speakers from various countries. As shown in Fig. 1(c), the learner compares the differences between the red parts of the model and the red parts of the speech shown by the system and repeats the speech in order to speak well (i.e., to make the red parts disappear).

As for distance, the native speaker's and the learner's utterances are represented by points on the two-dimensional coordinates in the diagram. The red dots represent the native speaker's speech and the blue dots represent the learner's speech. The greater the distance between these points, the more the user's utterance differs from the native speaker's utterance. As shown in Fig. 1(c), the user changes his pronunciation so that the distance between these points is closer. By doing so, it is expected that the user will be able to produce speech that is closer to the speech of the group for which the learning is intended. Now, let me explain why we need distance visualization as well as difference visualization. Visualization of differences mainly plays the role of informing the user of where the differences are. By evaluating the detailed aspects such as intonation, rhythm, phonemes, etc., we can get some idea of how the user's speech has changed. However, the proposed system does not use such metrics because learners do not necessarily improve their intelligibility by correcting their own speech in these aspects. Then, it is necessary to determine from the raw waveform diagram alone how the pronunciation changed when it was changed. However, the raw waveform alone does not provide enough information, and it is difficult for the average user to make a judgment based on this alone. Therefore, the user's speech and the native speaker's speech are each represented by a point on two dimensions, and the user can understand the change in the speech by visually seeing the trajectory of his or her point. For example, if the user's point moves closer to the native speaker's point along a certain arc every time the user repeats an utterance, then the pronunciation change is correct. On the other hand, if the point suddenly moves to a strange place when the user changes his or her speech, it suggests that the user may be modifying his or her speech in the wrong way.

3 IMPLEMENTATION

Our system uses deep learning techniques to identify the speech of native speakers and learners. This task is not very difficult if we can collect a large amount of data from each. However, it is usually very difficult to collect speech data of learners' utterances. Therefore, we apply self-supervised learning so that even a small amount of data can be used for sufficient discrimination. We use wav2vec 2.0 [1] as a self-supervised learning model, and the features obtained by wav2vec 2.0 are input to Bi-LSTM [5, 6] for classification. As a prototype, when we trained the model using data from Japanese learners of American English and data from American English speakers, the classification accuracy of the test data was 0.99 on the F-measure. In addition, we use the attention mechanism [3, 7] to visualize the difference between the native speaker's speech and the user's speech. Attention is a technique that mimics cognitive attention, and this mechanism highlights the important aspects to be considered when discriminating between native speakers and learner data. We highlight the areas highlighted by this mechanism in red on the waveform to make it easier for the user to understand.

Furthermore, we use metric learning to visualize the distance between the native speaker’s speech and the user’s speech. Metric learning is the task of learning a distance function between objects. We use triplet loss, which is one of the main approaches. Triplet loss is trained on a set of three audio sources: an anchor audio source, a positive audio source of the same class as the anchor, and a negative audio source of a different class than the anchor. The distance from the anchor audio to the positive audio is minimized and the distance to the negative audio is maximized. In this method, for example, if the native speaker’s speech is the anchor speech in a given sentence to be studied, the positive speech is the speech of another native speaker of that sentence, and the negative speech is the speech of a non-native speaker of that sentence. Thus, the native speech and the native speech are learned to move closer together, the non-native speech and the non-native speech to move closer together, and the native speech and the non-native speech to move further apart. The distance thus created is used to visualize the distance between the native speaker’s speech and the user’s speech.

4 CONCLUSION AND FUTURE WORK

In this paper, we present a novel system to assist language learners. This system not only judges the user’s speaking skill, but also shows

where the user differs from native speakers or model speakers, and how much the user differs from native speakers or model speakers. We expect that the user’s speech will gradually become better through interaction with this system. In the future, we plan to verify whether the user’s speech becomes better through user studies.

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